

F24-122-R-CricMIC AI Generated Commentary

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Chapter 1

Introduction

Cricket mobile gaming is prevalent however the gameplay experience lacks contextually aware commentary. Current games have preset commentaries that don't capture the whole context and diminish the excitement. Our project aims to provide a web interface where mobile cricket players will upload their gameplay video and receive contextually relevant commentary.

1.1 Motivation

The excitement of playing any mobile game is enhanced through good commentary but unfortunately, existing cricket games don't engage the players due to pre-recorded commentaries. The kind of commentary these games provide disconnects the players and they don't enjoy their gameplay. Additionally, few commentary voices are available which hinders the players from adding commentary of a new person of their choice.

1.2 Problem Domain

The current state of commentary in cricket mobile games is static and lacks contextual relevance, with pre-recorded audio that fails to reflect in-game actions dynamically. This disconnects players from their gameplay experience, reducing engagement and excitement. Moreover, the limited availability of commentary voices restricts customization, preventing players from choosing new voices to personalize their experience. There is a need for a solution that delivers real-time, context-aware commentary tailored to each gameplay scenario.

1.3 Research Problem Statement

Current mobile games deliver preset commentaries that are available in limited commentators' voices. These commentaries are not only repetitive but they are also unable to capture the complete context of the game. We will provide a user-friendly web interface that will handle commentary generation that will be non-repetitive and contextually aware. This will increase audience engagement who will benefit from informed commentary.

1.4 Software Requirements Specification

1.4.1 Functional Requirements

- **User Authentication:** The software should support secure login and account management, allowing users to access personalized settings and stored data.
- **Video Upload and Processing:** Users must be able to upload cricket match clips, which will then be trimmed, analyzed, and processed for event detection, scoreboard extraction, and commentary generation.
- **Preset Selection:** The system should allow users to choose predefined commentary styles and configurations to suit different commentary tones.
- **Voice Selection:** Users should be able to select from a variety of voices for the generated commentary, with options for accents, genders, and tones.
- **Real-Time Processing:** The software should provide commentary generation in real-time or near real-time, based on the uploaded video clips.

1.4.2 Non-Functional Requirements

These define the operational qualities and constraints of the software, ensuring performance, usability, and scalability:

- **Performance:** The system should be able to handle multiple video uploads simultaneously and process them efficiently without significant delays.
- **Usability:** The interface must be intuitive, enabling users to easily navigate the software and make selections for uploads, presets, and voices.
- **Scalability:** The system must be capable of scaling to accommodate an increasing number of users and video uploads as demand grows.

- **Security:** Strong security measures must be in place to protect user data, including encrypted storage of login credentials and safe handling of uploaded files.
- **Compatibility:** The software should be compatible across various browsers and devices, ensuring accessibility for all users.

Chapter 2

Literature Review

This chapter critically examines existing literature relevant to the CricMic’ architecture. It identifies key studies, theories, and findings, providing a foundation for CricMic and highlighting its relationship to prior work.

2.1 Related Research

Mentioned Down Below is the research we were able to gather related to our project.

2.1.1 MatchTime: Towards Automatic Soccer Game Commentary Generation

In recent years, significant work has been conducted to improve automatic soccer commentary generation, with datasets like SoccerNet-Caption serving as the foundation for many models. This paper proposed MatchTime, a curated dataset that corrects these temporal inconsistencies. MatchTime is built using a two-stage multi-modal temporal alignment pipeline that filters and adjusts the timestamp discrepancies in existing datasets. By aligning the visual content with the commentary text more precisely, MatchTime enables the training of models that generate more accurate and engaging soccer game commentary.

2.1.1.1 Summary of the research item

This paper focuses on the challenge of video-text misalignment in soccer commentary datasets, specifically SoccerNet-Caption, which collects commentary from over 400 matches. The study highlights a significant issue where commentary timing can lag behind game

events by up to 152 seconds, affecting the accuracy of AI-generated commentaries. To address this, the researchers manually corrected the timestamps for part of the dataset, creating the SN-Caption-test-align benchmark. This adjustment led to notable improvements in AI commentary accuracy, demonstrating the critical role of temporal alignment in enhancing commentary quality. The study suggests a multi-modal pipeline to automate this alignment for larger datasets.

2.1.1.2 Critical analysis of the research item

The commentary model leverages a range of visual features such as C3D, ResNet, Baidu, CLIP, and InternVideo, with features extracted at 1 or 2 FPS depending on the method. A fixed number of 32 query vectors are used in the temporal aggregator, and the aggregated features are projected to a 768-dimensional token that feeds into the LLaMA-3 model for text generation. The learning rate is set at $1 \times 10 \exp(-4)$, with training conducted over 100 epochs using a single Nvidia RTX A100 GPU. The study employs several popular metrics for evaluating commentary quality, including BLEU, METEOR, ROUGE-L, CIDEr, and the GPT-score, which assesses semantic accuracy, expression, and professionalism.

Method	Visual Features	BLEU-1	BLEU-4	METEOR	ROUGE-L	CIDEr	GPT-score
Trained on original SoccerNet-Caption							
SN-Caption	C3D	22.13	4.25	23.14	23.25	11.97	5.80
	ResNet	26.46	5.33	23.58	23.58	13.71	6.28
	Baidu	<u>29.61</u>	<u>6.83</u>	25.38	25.28	20.61	6.72
MatchVoice (Ours)	C3D	28.85	5.62	23.29	26.69	19.06	<u>6.90</u>
	ResNet	28.75	5.87	23.78	26.69	20.65	6.75
	InternVideo	28.50	6.24	24.30	30.75	23.34	6.80
	CLIP	28.65	6.62	24.20	27.33	<u>24.35</u>	6.78
	Baidu	30.32	8.45	<u>25.25</u>	<u>29.40</u>	33.84	7.07
Trained on our aligned MatchTime							
SN-Caption	C3D	26.81	5.24	23.57	23.12	13.78	6.27
	ResNet	27.63	5.75	24.05	23.42	15.65	6.33
	Baidu	<u>29.74</u>	<u>7.31</u>	26.40	26.19	23.74	6.84
MatchVoice (Ours)	C3D	28.67	6.55	24.46	27.38	26.53	6.89
	ResNet	29.21	6.60	24.11	24.32	28.56	6.84
	InternVideo	29.18	6.89	25.04	28.18	<u>30.22</u>	<u>6.99</u>
	CLIP	29.56	6.90	24.62	31.25	28.66	6.82
	Baidu	31.42	8.92	<u>26.12</u>	<u>29.66</u>	38.42	7.08

Figure 2.1: Evaluation Metrics

2.1.1.3 Relationship to the proposed research work

This paper uses Vision Transformers (ViTs), Perceiver-based Attention modules, and LLMs for generating text. The methods can be used in our problem to develop enhanced, context-based cricket commentary. We will use the idea of using a prompt to generate a text using any LLM like ChatGPT, Llama, or Claude. This approach is better than training a separate text-to-text model for commentary generation. Another way we can use the concepts in this paper is for Cricket Shots Classification, the ViTs can be used as a fea-

ture extractor of our Cricket Shots Classification model. The methodology of extracting features from videos can be useful in developing our classification model.

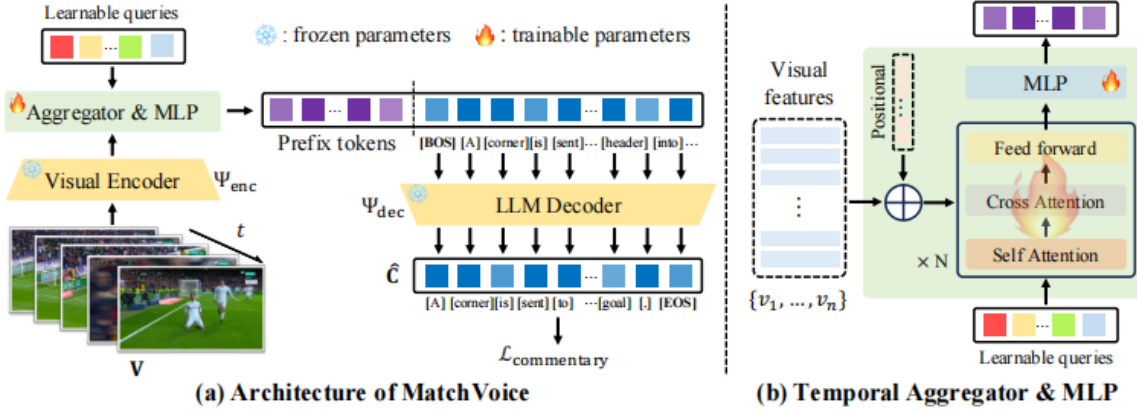


Figure 2.2: MatchTime: Towards Automatic Soccer Game Commentary Generation

2.1.2 CricShotClassify: An Approach to Classifying Batting Shots from Cricket Videos Using a Convolutional Neural Network and Gated Recurrent Unit

This paper focuses on the challenging task of classifying cricket batting shots using deep learning techniques. Traditional methods of manual feature extraction from video frames are time-consuming and prone to errors due to the high similarity between different cricket shots. This paper introduces a hybrid deep learning model combining a Convolutional Neural Network (CNN) and a Gated Recurrent Unit (GRU) to classify 10 distinct cricket batting shots, from cover drives to pull shots, using the CricShot10 dataset. By leveraging automatic feature extraction through CNN and addressing temporal dependencies via GRU, the model achieves a high classification accuracy. The study further investigates various transfer learning models, with the VGG16–GRU architecture demonstrating superior performance, providing a significant contribution to the field of cricket video analysis.

2.1.2.1 Summary of the research item

This paper introduces a novel method for classifying 10 different cricket batting shots using a hybrid deep-learning model. A new dataset, CricShot10, is created, containing videos with varying lengths and lighting conditions. The authors develop a CNN–GRU architecture and explore transfer learning models like VGG16, InceptionV3, Xception, and DenseNet169. By freezing layers in these models, the VGG16–GRU combination achieves the highest shot classification accuracy of 93. The study demonstrates how ef-

fectively CNNs extract spatial features, while GRUs capture temporal patterns, addressing the challenges posed by similar cricket shots and varying video conditions.

2.1.2.2 Critical analysis of the research item

The CricShot10 dataset is a valuable resource for future research, enabling benchmarking of models in cricket shot classification. The use of hybrid CNN-GRU architectures is well-suited for capturing the spatial and temporal aspects of sports videos. However, the dataset's limited size and shot variety pose a constraint. Expanding it to include more complex shots, like reverse sweeps, could better assess the model's generalization. While VGG16-GRU performs well, exploring advanced transformer-based models may offer further improvements in sequence modeling. Despite these limitations, the study offers a solid approach with potential applications in cricket coaching, commentary, and commercial use.

Name	Training Set	Test Set
Cover Drive	153	35
Defensive Shot	157	35
Flick	146	35
Hook	146	35
Late Cut	147	35
Lofted Shot	151	35
Pull	144	35
Square Cut	160	35
Straight Drive	154	35
Sweep	159	35

Figure 2.3: Dataset video quantity for 10 cricket batting shots

Model	Trainable Parameters	Precision (%)	Recall (%)	F1 Score (%)
CNN-GRU Model	13,802,134	82.30	81.70	82.00
DCNN-GRU Model	18,419,670	83.00	82.60	83.00

Figure 2.4: Parameters and various accuracy measures for custom models

2.1.2.3 Relationship to the proposed research work

One major task in our project is cricket shots (pull shot, cover drive, etc.) classification. This paper provided a dataset that we can use to build a main model and then fine-tune the resultant model on our dataset. The models developed in this paper can also give a good baseline for the model we will use in our project.

2.1.3 Designing for Automated Sports Commentary Systems

This paper explores the integration of Natural Language Processing (NLP) and Computer Vision (CV) to revolutionize automated sports commentary. Traditional sports commentary enhances viewer engagement, but with advancements in AI, the potential for automated, real-time commentary is gaining attention. The proposed system combines text-to-speech with visual cues for play-by-play and color commentary, creating a personalized and engaging experience for users. By leveraging CV for in-game analysis and NLP for generating commentary, this system offers a comprehensive solution for both auditory and visual feedback.

2.1.3.1 Summary of the research item

The research presents an automated sports commentary system that integrates Natural Language Processing (NLP) and Computer Vision (CV) to provide a dynamic and immersive experience. By using CV to detect in-game events in real-time, the system generates play-by-play commentary that narrates the ongoing action. This is further enriched with color commentary, which pulls additional contextual information from a database, such as player stats or background stories, to provide deeper insights into the game.

In addition to auditory feedback, the system enhances viewer engagement by incorporating visual elements like italicized text, which draws attention to key moments and improves overall immersion. This multimodal design ensures that the audience receives a more comprehensive and engaging commentary experience, blending both audio and visual cues. A user study conducted with eight participants sheds light on how viewers

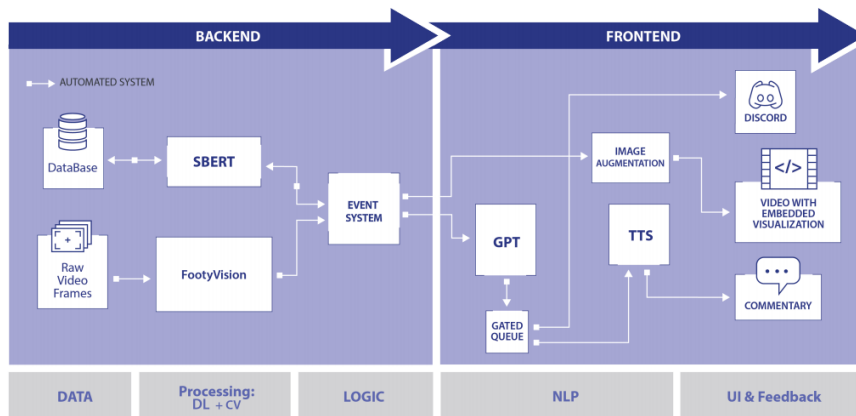


Figure 2.5: Designing for Automated Sports Commentary Systems

respond to this automated system. It emphasizes the need for personalized commentary content that aligns with different audience preferences, further validating the system's ability to enhance the sports broadcasting experience.

2.1.3.2 Critical analysis of the research item

While the system effectively combines NLP and CV to deliver automated sports commentary, one limitation is its reliance on traditional deep learning architectures. Although the current approach performs well in integrating real-time commentary and contextual analysis, further experimentation with transformer-based models could enhance language precision and adaptability. Additionally, the relatively small sample size of the user study may not fully represent the diversity of audience preferences, warranting further research with a larger participant pool. Nonetheless, the research lays a strong foundation for future developments in personalized and AI-powered sports commentary systems. The Implementation was performed on MSI laptop with a 12th Gen Intel I9-12900HK 2.90Ghz CPU, NVIDIA GeForce RTX 3080 Ti GPU, and 32GB Ram.

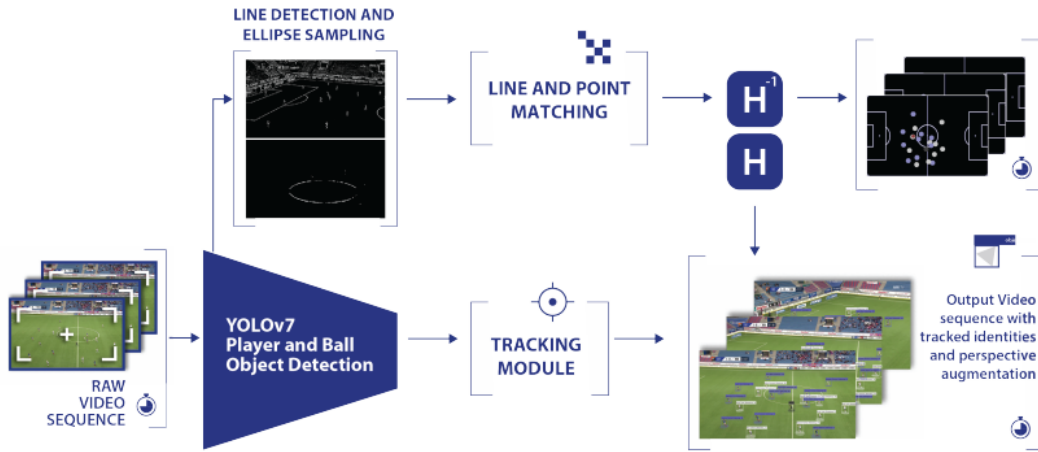


Figure 2.6: Critical Analysis

2.1.3.3 Relationship to the proposed research work

The purpose of this paper is to provide a holistic picture of designing a sports commentary system, and the challenges and solutions suggested in this paper will be vital for designing our cricket commentary generation application. An additional mechanism adopted in this paper is to include relevant data from external sources during the commentary generation process to create a color commentary system. This task can be included in our project when we are using the text generation models. We can incorporate external data relevant to the players, teams or matches involved in the video. It provides an extensive list of models available for video classification, text generation, and speech generation. The speech models mentioned in the paper will be useful for us in generating dynamic commentary.

2.1.4 Optimized deep learning-based cricket activity

Activity recognition in sports plays a crucial role in analyzing the performance of individual players and teams across different athletic events. Each sport has distinct dynamics, from simple activities like walking and sitting to more complex actions such as overhead smashes in tennis. This paper introduces a new cricket activity dataset and a deep learning model that effectively handles the complex task of activity recognition in cricket. By combining CNNs and LSTMs, the proposed model captures both spatial and temporal features, outperforming existing methods on both the new cricket dataset and four standard benchmarks.

2.1.4.1 Summary of the research item

The proposed model consists of multiple stages, beginning with keyframe selection and resizing to a standard size of 100x100 pixels for efficient training. The CNN layers are used to extract spatial features from the resized keyframes. These spatial features are then passed through LSTM cells to capture temporal dependencies between frames. The CNN component comprises multiple convolutional layers, each with a set of filters that detect patterns in the input keyframes. The output of these layers is a set of high-level feature maps that capture important details relevant to activity recognition. The LSTM component then analyzes these feature maps to identify the sequence of actions occurring in the video.

2.1.4.2 Critical analysis of the research item

The authors have conducted extensive experiments on the newly created cricket dataset as well as four benchmark datasets: UCF101, HMDB51, YouTube Action, and Kinetics. The experiments were carried out using Keras with TensorFlow-GPU on a system equipped with an NVIDIA GeForce GTX 3060 GPU. The results indicate that their model significantly outperforms other RNN-based approaches in terms of accuracy, recall, and F1 score.

2.1.4.3 Relationship to the proposed research work

This paper also introduces a medium-scale dataset that includes five cricket activities that will be used in our project. The authors have experimented with various vision encoders with multiple RNN-based decoders to produce an efficient model. We are planning on using similar methods on our dataset.

2.1.5 Towards Automated Video Game Commentary Generation using Generative AI

Video gaming is growing as an industry and it provides unique challenges which could be solved using AI. One of the popular use cases of video games has been in the field of Reinforcement Learning where the effectiveness of Agents is determined by how well they perform on the chosen game. With the advent of Generative AI, the authors of this paper have tried to incorporate deep learning models and APIs to generate dynamic commentary of League of the Legends video game.

2.1.5.1 Summary of the research item

The authors have used pre-trained CNN models to extract frame-level features of size 1028 which is then passed to multi-layer perceptron for classifying the video clip into one of the 20 classes. Then a simple matching dictionary is created that assigns a text to each class which along with a pre-defined prompt is sent to the LLM, to generate commentary. The generated is sent to a FakeYou API for generating the speech. The whole process is very effective in generating commentary, as everytime a unique commentary is generated for the same action. The authors have also incorporated a priority queue mechanism to determine which events should be used to generate commentary. This ensures that there is commentary generated every 15-20 seconds.

2.1.5.2 Critical analysis of the research item

Although the authors have utilized state-of-the-art models for classifying events, generating text, and converting the text into speech, they fail in capturing the complete scene of an event as combat games provide very challenging scenes to be understood by the machine. Additionally, they are working on processing one frame every second which provides very little information as the temporal information has not been utilized. Thirdly, the generated commentary is very generic and no mechanisms have been adopted to tackle the problem of hallucination. Therefore, despite providing useful techniques this paper has some weaknesses that need to be catered to when adopted in our project.

2.1.5.3 Relationship to the proposed research work

This paper closely matches with our project as it includes end-to-end commentary generation. It includes modules like video classification, text generation, and voice generation. We also have to work on such tasks in order to generate cricket game commentary. This paper uses CNN based models like ResNet and EfficientNet to capture video frames' fea-

tures and MLP for accurately classifying an activity. Once again the models will be used in our project for cricket shots classification and the APIs mentioned in the paper could be tested for generating text and speech.

2.1.6 Automatic Cricket Highlight generation using Event-Driven and Excitement-Based features

Generating sports highlights is a labor-intensive task that traditionally requires a certain level of expertise. With a special focus on cricket, a sport known for its complex rules and extended playing time, the authors propose an automated framework for generating sports highlights. Cricket matches include numerous key moments such as boundaries, wickets, and milestones, and the proposed model incorporates both event-driven and excitement-driven elements to capture important moments from the match. Key features like replays, audio intensity, player celebrations, and match scenarios are used to detect and extract crucial highlights.

2.1.6.1 Summary of the research item

The presented model operates in a structured manner, aiming to break down a cricket match's footage into essential segments. This is accomplished by dividing the match into overs, further breaking them down into individual video shots. Each of these shots represents an important aspect of the match such as player actions, boundary hits, and wickets. The process of highlight generation follows three stages: first, the model identifies various video shots within the innings (Stage 1); second, it extracts relevant cues like player celebrations, scoreboard updates, and audio cues (Stage 2); and finally, it produces highlights based on either event-driven or excitement-driven criteria (Stage 3).

2.1.6.2 Critical analysis of the research item

This model addresses a significant challenge in sports broadcasting: automating the process of highlight generation, particularly for a complex sport like cricket. The approach of combining both event-driven (boundaries, wickets, etc.) and excitement-driven cues (replays, audio intensity, player celebrations) is particularly commendable, as it ensures the highlights not only capture key moments but also the overall excitement and atmosphere of the match. By using such multimodal cues, the model aligns well with human viewing preferences, focusing on both the technical events and the emotional highs of the game.

2.1.6.3 Relationship to the proposed research work

This seminal paper published in CVPR in 2018 uses multiple methods for detecting key events in an untrimmed cricket video. The methods include detecting scoreboard, umpire gestures, audience voice analysis and CNN based activity recognition. Although the task performed in this paper, highlights generation, is different from ours but certain methods like scoreboard detection, player detection are the same. The methodology in this paper can be applied for detecting scores, players and later on in the shots classification model. Various frames sampling methods have been applied to work on key frames during the classification of an action which is quite useful and relevant to our work.

2.1.7 Goal: Event Detection in Sports

Understanding complex events from unstructured videos, such as scoring a goal in a football game, is a highly challenging task. This is primarily due to the dynamic, complex, and varied nature of video sequences. In this study, the authors tackle this problem by leveraging the power of deep learning frameworks. Specifically, they propose a method that independently encodes spatial and temporal information using convolutional neural networks (CNNs) and subsequently fuses these features using regularized autoencoders. The research is demonstrated on a newly compiled dataset consisting of goal and no-goal sequences.

2.1.7.1 Summary of the research item

This paper presents a novel method for understanding and detecting complex events in unstructured video sequences, particularly focusing on football matches. The authors propose a two-step approach: (i) extraction of spatio-temporal features using CNNs and (ii) fusion of these features via regularized autoencoders to perform classification of goal and no-goal events. The model effectively handles the dynamic challenges of sports videos, such as varying vantage points and zoom levels, by encoding both spatial and temporal data separately before combining them for classification.

2.1.7.2 Critical analysis of the research item

This research makes a significant contribution to the field of sports video analytics by addressing the complexities associated with unstructured, dynamic video streams. The separation of spatial and temporal information before feature fusion is a clever approach, as it tackles the challenges posed by shifting perspectives and changing camera zooms in sports broadcasts. The use of regularized autoencoders for feature fusion further strength-

ens the model’s ability to generalize across varying video conditions. However, the research could benefit from a more comprehensive exploration of larger and more diverse datasets, as the current model’s success is largely demonstrated on a limited dataset. Additionally, more insights into how well the model performs in non-football sports or other video contexts would add to the generalizability of the approach.

2.1.7.3 Relationship to the proposed research work

One of the primary tasks in our project is to detect an event in a video like a fall of a wicket or boundary scored. The methods described in this paper can be used to detect major events in a video. This uses the combination of classical methods like HOG and SIFT with deep learning models like CNN to detect an event in a sequence of frames. The model architecture can be implemented for detecting events as well as in classifying the cricket shots. Although the models used are very simple, they can still give us a baseline on evaluating the performance of different models.

2.2 Analysis Summary of research items

Commentary generation has been tried on many sports games including soccer, baseball, cricket, and other video games such as League of the Legends [1].

The following table shows papers that have been recently published to address different aspects of commentary generation.

Ref No.	Methodology	Dataset
[1]	Time Distributed 2D CNN, LSTM	Cricket Shots (Custom Dataset)
[3]	Visual Encoders, Perceiver, LLaMa	SoccerNet-Captions
[4]	Vision Transformers, Prompt Narration	Custom Dataset
[5]	CNN-GRU, Transfer Learning (VGG16, InceptionV3, DenseNet169)	CricShot10
[2]	Multimodal System (CV, NLP Integration)	Real-Time Sports Commentary (Custom)

Table 2.1: Review of Papers, Methodologies, and Datasets

Each paper gave us a new insight on how to approach this problem. In [3] and [4] LLMs were used for commentary generation. [3] also presents the common evaluation metrics

(BLEU, METEOR, ROUGE, etc.) used for text-generation tasks. Furthermore, [1] presents an architecture for shot classification, and [5] uses YOLO for player detection.

Chapter 3

Proposed Approach

To generate dynamic cricket commentary we have to implement various Computer Vision, Natural Language Processing, and Speech Generation models. The proposed method involves extracting scoreboard information through OCR, player detection, shot classification, commentary generation through a Large Language Model, and text-to-speech conversion for audio generation.

3.1 Pre-Processing

The pre-processing stage focuses on preparing raw video footage for subsequent analysis. In this step, cricket clips are trimmed to isolate specific events, such as when a ball is bowled and the batsman plays a shot. By trimming clips, we ensure that only relevant portions of the video are passed through the processing pipeline, reducing computational overhead and improving the accuracy of downstream models.

3.1.1 Clip Trimming

During the test phase, the first step is to trim the incoming video clips based on cricket-specific events. We utilize heuristics and predefined rules to identify key moments in the game, such as the start of a bowler's run-up, the point when the ball is delivered, and when the batsman makes contact with the ball. These segments form individual "events," and the video is trimmed accordingly to include only the footage around these moments.

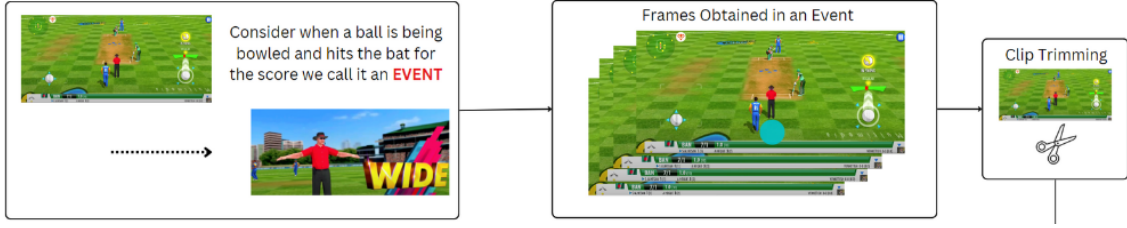


Figure 3.1: Clip Trimming

3.1.2 Bounding Box Detection

To accurately detect the scoreboard in cricket videos, we manually annotated data to mark where scorecards typically appear. Using this annotated data, we trained a model to detect a bounding box around the scoreboard in each frame. This helps localize key information like scores and overs.

By identifying the scoreboard's position, we ensure that it can be reliably extracted for further processing, such as Optical Character Recognition (OCR). This step strengthens the model's robustness across different camera angles and lighting conditions, allowing the system to consistently capture scoreboard information for commentary generation.

3.2 Event Detection

The current context of the game, which we will call an event, can be extracted through the scoreboard. The scoreboard provides various information such as runs, wickets, current run rate, required run rate, etc. This information could be extracted using OCR models.

3.3 Players Detection

Similar to score extraction, player information is also extracted through the help of OCR models. This will help in constructing a prompt which will be sent to a LLM for generating a commentary.

3.4 Shot Classification

To encapsulate additional context we will classify the shots played by the player such as cover drive, pull shot, straight drive, etc. This can be done by fine-tuning a Video Classification model on our dataset. We utilize an additional dataset from Kaggle that

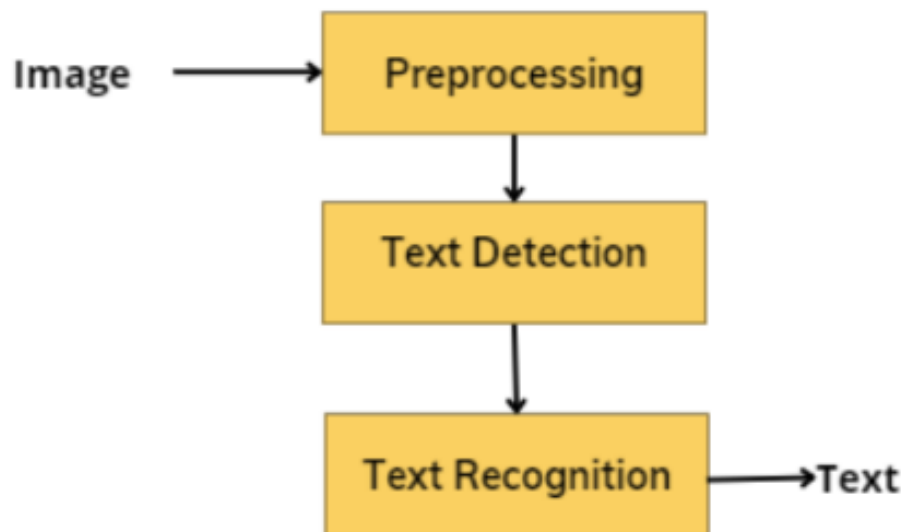


Figure 3.2: Event Detection

contains annotations for various cricket shots. This dataset is integrated to further enhance the model's ability to detect and classify different types of shots played during the match, contributing to more accurate and context-aware commentary generation. This is then added to already acquired information for commentary generation.

3.5 Audio Captioning

Some information could not be present in the scoreboard which can be extracted through the present commentary which can be added to the prompt to enhance the context.

3.6 Text-to-Text

Once all the information is available, an LLM could be used to generate the commentary through a prompt. This commentary will contain the whole context of the game. Each time it would be an enhanced and unique commentary.

3.7 Text-to-Speech

Finally, once an enhanced, contextually-aware commentary is generated it can be sent to a speech model to generate audio. To make the commentary realistic, it should have inflection in voice to capture various emotions. The text-to-speech models will be capable of producing such audio. To limit our scope we have decided to use a single commentator's voice for commentary generation. We would need to fine-tune models on the selected commentator's voice.

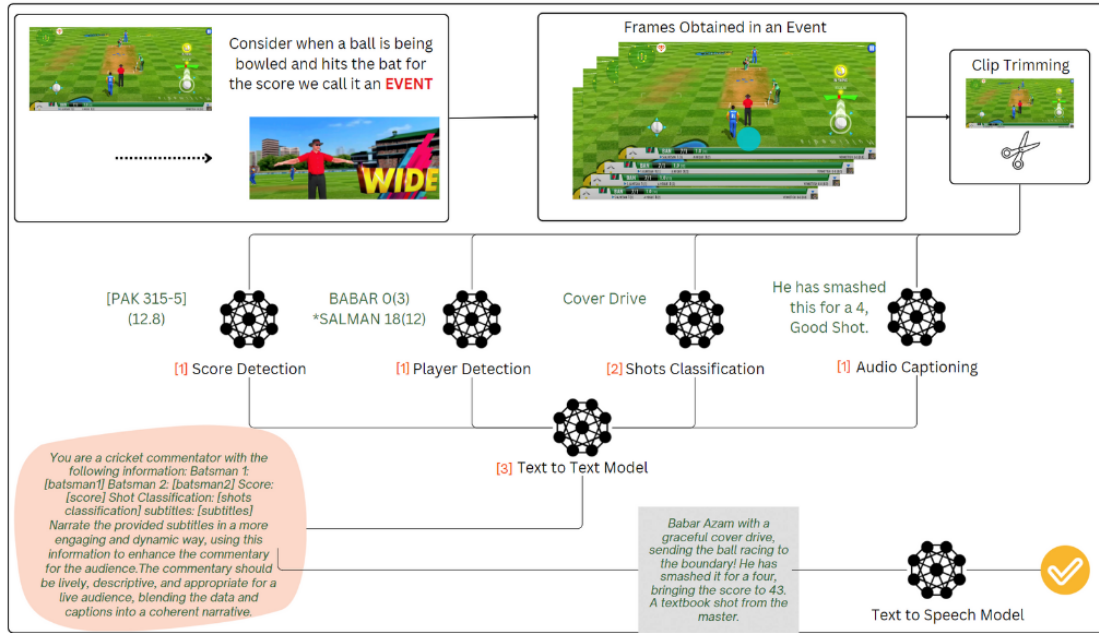


Figure 3.3: Architecture Diagram

3.8 Web Application

The web application serves as the user interface for generating dynamic cricket commentary, providing an intuitive platform to interact with the system's features. The application is designed with ease of use in mind, allowing users to upload videos, select presets, and customize voices for the generated commentary. Key features of the web application include:

3.8.1 Login

Users are required to log in to the application to access its functionality. The login system ensures that each user's data is secure, and it provides a personalized experience, allowing

users to save presets and voice preferences for future use. Once logged in, users can navigate through the platform seamlessly.

3.8.2 Sidebar

The sidebar offers easy navigation within the application, providing access to core functionalities such as uploading videos, selecting presets, and choosing voice options for commentary. It remains accessible on every page, allowing users to interact with multiple features simultaneously.

Upload: The upload feature allows users to input their cricket videos directly into the system. Once uploaded, the application processes the video, trims it into events, and analyzes it using the pre-trained models for player detection, shot classification, and scoreboard extraction.

Presets: Users can choose from various presets that define how the commentary is generated. These presets include different commentary styles, such as professional, casual, or enthusiastic, tailoring the tone and language of the output based on user preferences.

Voices: This feature enables users to select from a wide range of voices for the commentary. Users can choose a default voice or personalize the commentary by selecting a specific commentator's style, gender, or accent. This flexibility enhances the user experience by creating commentary that feels authentic and aligned with their preferences.

3.9 Diagrams

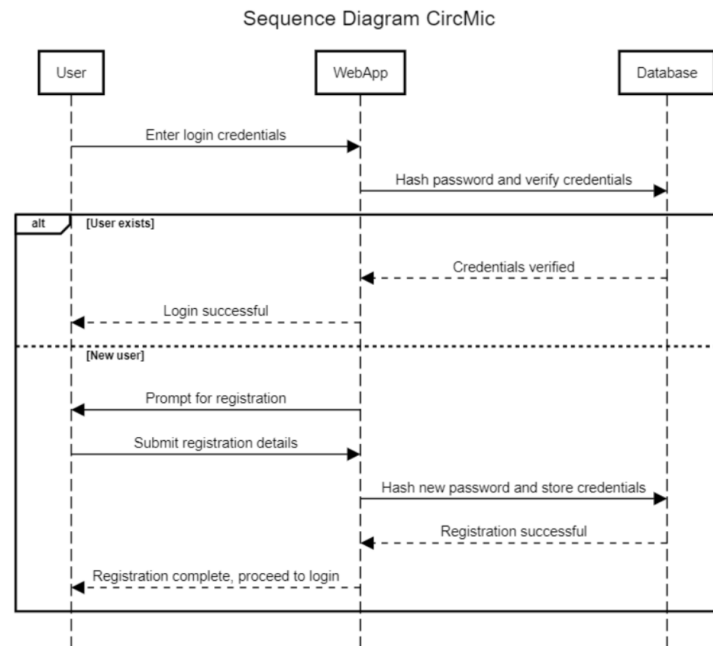


Figure 3.4: Login

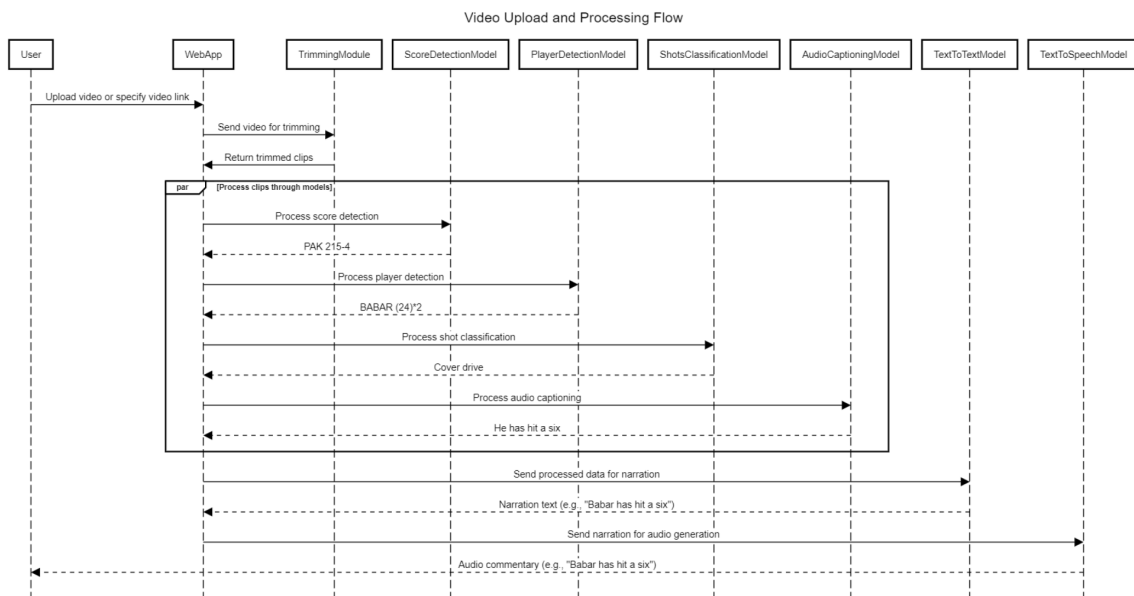


Figure 3.5: Models Functionality

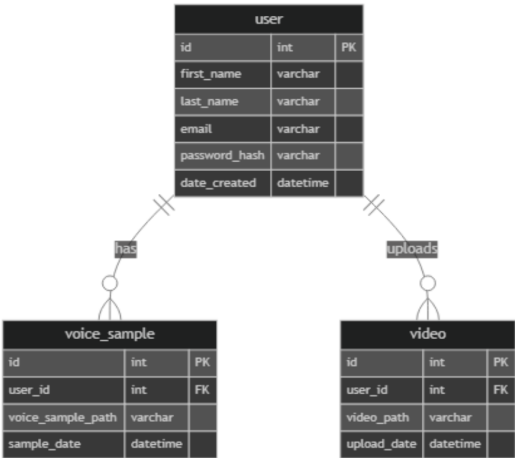


Figure 3.6: ERD

ooo

Login

Sign in to continue

EMAIL

hello@reallygreatsite.com

PASSWORD

login

Figure 3.7: Mock Fig 01

ooo

Register

Sign Up to continue

FIRST NAME

James

LAST NAME

Ranch

EMAIL

james@gmail.com

PASSWORD

USERNAME

james@gmail.com

Register

Figure 3.8: Mock Fig 02

ooo

CricMic

Generate

Presets

Voices

Preference

My Profile

Settings

Support

Help & Support

Match Coverage

SL vs NZ
2nd Men's Test

Champions Cup
Pak Domestic

Ind vs Ban
Men's Test series

CPL 2024
T20 league

Eng vs Aus
3rd Men's ODI

Ahmad Shah Abdali
Afghanistan domestic



Live - Mathews gets a life after Karunaratne run-out

1st Inn SL 158/2 (47.2 ov) vs: NZ

- Lunch: Century stand for Chandimal-Karunaratne
- NZ, SL wary of another collapse in Galle conditions
- Jayasuriya moves up in ICC bowling rankings
- Versatile SL inch up WTC table in contrasting conditions

WTC points table

1st Test: Sri Lanka won by 63 runs

Top Stories



Miller on T20 WC loss: 'I probably haven't gotten over it yet'

6 hrs ago • Interview by Nagraj Gollapudi and Deivanayan Muthu



Ind vs Ban: Shakib 'eligible for selection' for Kanpur Test

22 hrs ago • Alagappan Muthu

See all

Figure 3.9: Mock Fig 03

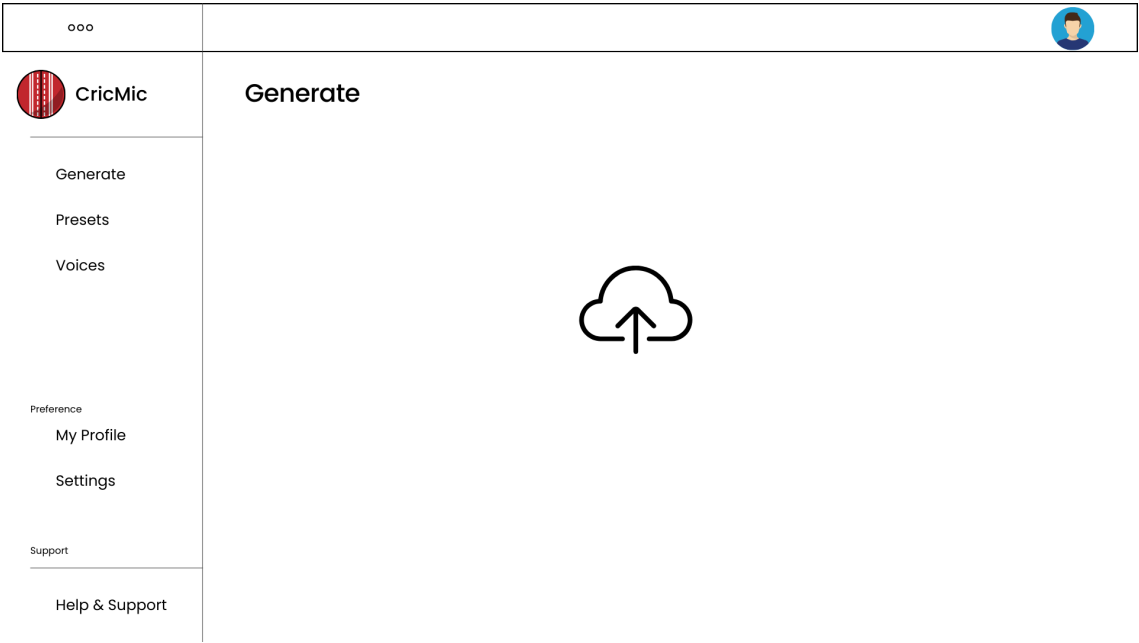


Figure 3.10: Mock Fig 04

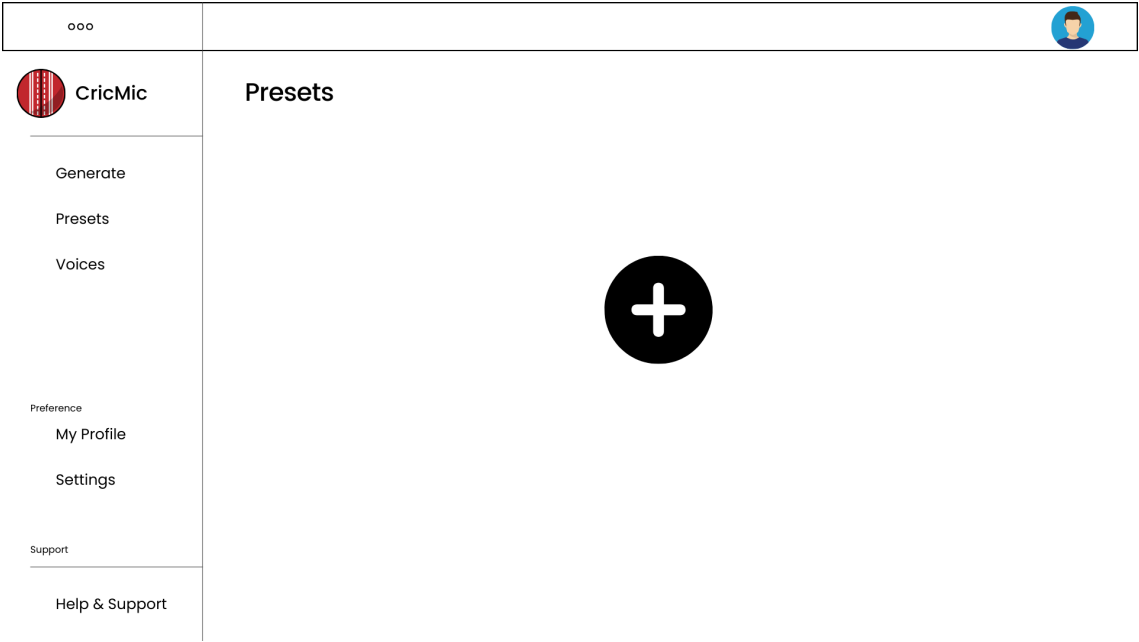


Figure 3.11: Mock Fig 05

3. Proposed Approach



Figure 3.12: Mock Fig 06

Test Case ID	BU_001	Test Case Description	Test the Login Functionality in CricMic			
Created By		Reviewed By	-		Version	2.1
QA Tester's Log	Test Case Passed!					
Tester's Name		Date Tested	24-Sep-2024		Test Case (Pass/Fail/Not Executed)	Pass
S #	Prerequisites:		S #	Test Data		
1	Access to Chrome Browser		1	Username = rafaykek		
2			2	Pass = df12@434c		
3			3			
4			4			
Test Scenario	Verify on entering valid username and password, the customer can login					
Step #	Step Details	Expected Results	Actual Results		Pass / Fail / Not executed / Suspended	
1	Navigate to HTTP://local-host:4200/login	Site should open				
2	Enter username& Password	Credential can be entered				
3	Click Submit	Customer is logged in				
4						

Figure 3.13: Login Test Case

Test Case ID	BU_002	Test Case Description	Test the UPLOAD Video Functionality in CricMic		
Created By		Reviewed By	-	Version	2.1
QA Tester's Log	Test Case Passed!				
Tester's Name		Date Tested	24-Sep-2024	Test Case (Pass/Fail/Not Executed)	Pass
S #	Prerequisites:		S #	Test Data	
1	Access to Chrome Browser		1	Video file	
2	Login with a verified account.		2		
3			3		
4			4		
Test Scenario	Verify on entering valid username and password, the customer can login				
Step #	Step Details	Expected Results	Actual Results	Pass / Fail / Not executed / Suspended	
1	Navigate to HTTP://local-host:4200/generate	Site should open			
2	Upload video	Video should be uploaded			
3					
4					

Figure 3.14: Uploading Test Case

Test Case ID	BU_002	Test Case Description	Test the Event Detection Functionality in CricMic		
Created By		Reviewed By	-	Version	2.1
QA Tester's Log	Test Case Passed!				
Tester's Name		Date Tested	24-Sep-2024	Test Case (Pass/Fail/Not Executed)	Pass
S #	Prerequisites:		S #	Test Data	
1	Access to Chrome Browser		1	Validation dataset	
2	Login with a verified account.		2		
3			3		
4			4		
Test Scenario	Verify on entering valid username and password, the customer can login				
Step #	Step Details	Expected Results	Actual Results	Pass / Fail / Not executed / Suspended	
1	Navigate to HTTP://local-host:4200/generate	Site should open			
2	Upload video	Video should be uploaded			
3	Download Video	Generated Video should be downloaded			
4	Measure the test data accuracy on the back end	Don't show the accuracy but keep it for versioning purposes			

Figure 3.15: Model Optimization

Chapter 4

Conclusions and Future Work

In conclusion, the engagement and excitement of any sport are highly dependent on how good the commentary is. The automatic generation of cricket commentary for mobile games will help in achieving this goal. We plan to provide a pipeline to generate automatic, context-aware, and dynamic cricket commentary. The commentary will be unique for each cricket event and will be non-repetitive. It will change according to the flow of the game and will present in-game insights. In addition, we will develop a complete web application for the gamers to upload their gameplay and receive enhanced commentary which could enhance audience engagement and retention.

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